

西安交通大学

博士学位论文

面向机载智能光电系统的实时目标感知算法研究与应用验证

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**Research and Application Evaluation of Real-Time Object
Perception Algorithms for Airborne Intelligent Electro-Optical
Systems**

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摘要

随着低空经济的快速发展，无人机作为其核心载体，在军事侦察、物流配送、公共安全等领域的应用日益广泛，相应地，对于无人机在自主环境感知与智能化方面的要求日益提升。机载光电系统是实现无人机智能化的关键，然而，在实际复杂动态的低空环境中，现有系统面临多重挑战。在算法功能层面，无人机独特的俯视视角导致目标尺度变化剧烈、在图像中像素占比低且常与复杂背景混杂，通用检测模型面对此类场景时性能急剧下降，航拍视角下目标频繁被建筑物、植被遮挡，现有跟踪器易丢失目标，难以保持稳定的长时跟踪。在系统层面，机载平台有严格的尺寸、重量、功耗约束，其信息处理核心必须采用高度集成化的嵌入式边缘计算模组，此类模组的算力与内存带宽与服务器相比存在数量级差距，仅能支持经过深度优化的轻量卷积神经网络运行。对于计算复杂度更高的基于 Transformer 的先进模型，现有边缘计算模组难以保证其高帧率实时处理，对于参数规模庞大的视觉-语言大模型（VLM），除了最新推出的高性能计算模组外，大部分的存量设备不具备部署此类模型的硬件基础。本文围绕“面向机载智能光电系统的实时目标感知算法研究与应用验证”这一主题，从工程落地角度出发，针对现有光电系统在实际使用中遇到的痛点难点，提供一套从核心算法到工程原型的系统性解决方案。在算法层面，分别设计面向可见光与红外图像的轻量化高精度目标检测模型，以及具备抗遮挡能力的长时单目标跟踪框架。在系统层面，设计并实现了一套集多光谱数据采集、嵌入式智能处理与实时通信于一体的工程原型系统，具体的研究工作如下。

1. 提出了面向可见光航拍图像的高性能小目标检测网络 BAP-DETR。针对无人机可见光航拍图像中目标尺度多变、背景复杂、小目标检测困难的问题，本文设计了双注意力处理模块（Bipartite Attentive Processing Block）与基于频域感知融合的双融合编码器（Dual-Fusion Encoder）。通过通道分离策略实现卷积与自注意力全局建模能力的优化，双融合编码器在融合高级语义信息的同时保留关键的低层特征，增强了对小目标特征的保留能力与多尺度特征的融合效果，并结合改进的损失函数在不增加模型推理计算复杂度的前提下提升了定位精度。在多个公开数据集上的实验表明，BAP-DETR 的平均检测精度与基线模型相比提升 6.9%，计算负载降低 17.5%，与最先进的可见光航拍图像检测器相比精度提升 3.8%，实现了精度与效率的平衡，为可见光航拍图像中的小目标检测提供了有效的解决方案。

2. 提出了面向无人机红外图像的轻量化小目标检测网络 MFF-DCNet。针对无人机红外图像小目标存在的低分辨率、低对比度、目标像素占比低及边缘设备算力受限等核心挑战，本文设计了深度可分离跨阶段 Transformer（Depth-wise Cross-stage Transformer, DCFormer）模块与多特征聚焦（Multi-Feature Focus, MFF）颈部结构，在确保高精度的同时，实现了在嵌入式平台上的实时推理。DCFormer 将深度可分离卷积与 Transformer

编码器结合,在降低计算复杂度的同时,增强了主干网络特征提取能力。多特征聚焦颈部结构通过构建全新的特征融合路径,有效聚合了不同层级的语义信息和空间细节。在 HIT-UAV 和 DroneVehicle 红外航拍数据集上的实验表明, MFF-DCNet 的检测精度与专用的无人机航拍图像目标检测器相比提升了 5.8%, FPS 提升了 10%, 与最新的 DETR 及其衍生方法相比, AP 至少提升 8.2%, FPS 是 RT-DETR 的 3 倍。同时, MFF-DCNet 在 Nvidia Jetson Orin NX 边缘模组上实现了 39.6 FPS 的实时性能, 展现出其在资源受限环境中的实际部署能力。

3. 提出了一种面向边缘计算设备的抗遮挡长时目标跟踪框架 SKF-Tracker。针对无人机在城市场景下面临的目标被复杂地物(如建筑、高架桥)遮挡所导致的跟踪失败的问题, 本文设计了一套可复用于任何基础跟踪器的抗遮挡单目标跟踪框架, 以结构相似性指数为基础构建跟踪置信评估与遮挡判定模块, 并结合基于卡尔曼滤波的轨迹预测以及动态模板更新策略, 实现复杂环境下的稳定目标跟踪。为覆盖多样化的城市场景, 本文在多个典型区域(如密集街区、交通枢纽、公园绿地)进行了大量实地无人机飞行测试, 采集原始视频素材。在此基础上, 从中筛选并构建了多模态无人机视角遮挡目标跟踪数据集 MMUOT-1050, 该数据集共包含 38 段高质量视频序列(20 段可见光, 18 段红外), 并对其中总计 1050 个目标实例(可见光 353 个, 红外 697 个)进行了逐帧的边界框标注, 每个目标都经历完整的遮挡后重新出现的过程。SKF-Tracker 在该数据集上实现了 89.37% 的可见光视频成功率和 91.93% 的红外视频成功率, 分别较基线方法提升 14% 和 11.73%。在实时性方面, SKF-Tracker 在保持高精度的同时, 仍能保持 31.25 FPS, 其速度远超基于深度神经网络的跟踪算法, 不占用端侧宝贵的 NPU 资源。为无人机在复杂环境中实现稳定长时监视任务提供了实用的解决方案。

4. 使用多种边缘计算模组设计并实现了机载智能光电图像处理分系统的工程原型。集多光谱数据采集、嵌入式智能处理与实时通信于一体, 以高集成度的嵌入式边缘计算模组作为信息处理核心, 支持多路可见光与红外视频流的采集与处理。在软件层面, 设计了一套高度模块化、可扩展的嵌入式软件框架, 框架采用多种设计模式, 实现了数据采集、处理与通信服务的高效协同, 通过引入无锁环形缓冲区、内存池与条件变量等机制, 提升了系统的实时性能。系统通过多种通信协议(TCP/IP, UDP, HTTP, 串口)与上位机建立双向数据通道, 实现了“人在回路”的混合协同架构。该方案在大疆 M350 RTK 无人机上完成了集成与实地飞行验证, 验证了其从数据采集、实时处理到信息交互的全链路工作能力, 这套方案不仅为本文的算法研究提供了可靠的工程验证载体, 也为后续算法部署与系统集成提供了具体的技术参考。

关键词: 无人机; 光电系统; 边缘计算; 目标检测跟踪

论文类型: 应用研究

ABSTRACT

With the rapid development of the low-altitude economy, drones, as its core platform, are increasingly used in military reconnaissance, logistics distribution, public security, and other fields. Consequently, the requirements for autonomous environmental perception and intelligent capabilities of drones are continuously rising. The airborne electro-optical system is crucial for enabling drone intelligence. However, in complex and dynamic low-altitude environments, existing systems face multiple challenges. In the aspect of the algorithmic and functionality, the unique top-down perspective of drones leads to drastic variations in object scale, low pixel occupancy in images, and frequent blending with complex backgrounds, causing the performance of general-purpose detection models to degrade sharply in such scenarios. Frequent occlusions by buildings or vegetation from an aerial view can easily cause existing trackers to lose targets, making sustained long-term tracking difficult. In the aspect of system implementation, airborne platforms are subject to strict size, weight, and power (SWaP) constraints. They must employ highly integrated embedded edge computing modules as their core processing units, whose computing power and memory bandwidth are orders of magnitude lower than those of server-grade GPUs. These modules can only support the deeply optimized lightweight convolutional neural networks. For more computationally complex advanced models based on the Transformer architecture, existing edge computing modules struggle to guarantee high frame rate real-time processing. Regarding large scale Vision-Language Models (VLMs), aside from the latest high performance computing modules, most existing devices lack the hardware foundation necessary for deploying such models. Focusing on the theme of "Research and Application Evaluation of Real-Time Object Perception Algorithms for Airborne Intelligent Electro-Optical Systems", this dissertation, from an engineering implementation perspective, addresses the difficulties encountered in practical applications of existing electro-optical systems by providing a systematic solution ranging from core algorithms to an engineering prototype. In terms of the algorithmic, lightweight high-precision object detection networks for visible-light and infrared images, as well as a long-term single object tracking framework with anti-occlusion capability, are designed respectively. In terms of system implementation, an integrated engineering prototype system for multi-spectral data acquisition, embedded intelligent processing, and real-time communication is designed and implemented. The specific research work is as follows.

1. This dissertation proposes a high-performance small object detection network BAP-DETR for visible-light aerial images. Addressing the challenges of significant scale variation, complex backgrounds, and difficulty in detecting small objects in drone-based visible-light images,

this work designs a Bipartite Attentive Processing Block and a Frequency-Aware Fusion Module based Dual Fusion Feature Encoder. A channel split strategy optimizes the global modeling capabilities of convolution and self-attention. The Dual Fusion Encoder fuses high-level semantic information while preserving critical low-level features, enhancing the retention of small object features and the integration of multi-scale features. Combined with an improved loss function, the localization accuracy is improved without increasing the computational complexity of model inference. Experiments on multiple public datasets show that BAP-DETR improves the average precision(AP) by 6.9% compared to the baseline while reducing the computational load by 17.5%, when compared to the state-of-the-art detector for UAV imagery the AP improved 3.8%, achieving a balance between accuracy and efficiency, thus providing an effective solution for small object detection in visible-light aerial images.

2. This dissertation proposes a lightweight small object detection network MFF-DCNet for drone infrared images. To address the core challenges in small object detection within drone infrared images, such as low resolution, low contrast, minimal object pixel occupancy, and limited computing power on edge devices, this dissertation designs a Depthwise Separable Cross-stage Transformer (DCFormer) module and a Multi-Feature Focus (MFF) neck structure. These designs enable real-time inference on embedded platforms while maintaining high accuracy. DCFormer combines depthwise separable convolution with a Transformer encoder to reduce computational complexity while enhancing the feature extraction capability of the backbone. The MFF neck constructs a novel feature fusion path to effectively aggregate semantic information and spatial details from different levels. Experiments on the HIT-UAV and DroneVehicle datasets show that MFF-DCNet improves detection precision (AP) by 5.8% compared to dedicated drone image detectors, with a 10% increase in FPS. Compared to the latest DETR and its variants, AP is improved by at least 8.2%, and the FPS is three times that of RT-DETR. Furthermore, MFF-DCNet achieves a real-time performance of 39.6 FPS on the Nvidia Jetson Orin NX, demonstrating its practical deployment capability in resource-constrained environments.

3. This dissertation proposes an anti-occlusion long-term object tracking framework SKF-Tracker for edge computing devices. Addressing the problem of tracking failure caused by occlusion from structures (e.g., buildings, overpasses) in urban drone scenarios, this dissertation designs a modular anti-occlusion single object tracking framework that can be integrated with any base tracker. It constructs a tracking confidence assessment and occlusion detection module based on the Structural Similarity Index Measure, combined with Kalman filter-based trajectory prediction and a dynamic template update strategy to achieve stable object tracking in complex environments. To cover diverse urban scenarios, this dissertation conducted exten-

sive drone flight tests in multiple typical areas, such as dense urban blocks, interchange, and parks to collect raw video footage. Based on this, a multi-modal UAV-perspective object tracking dataset MMUOT-1050 was constructed. This dataset comprises a total of 38 high-quality video sequences (20 visible-light and 18 infrared). Within these sequences, a total of 1,050 instances (353 in visible-light and 697 in infrared videos) were annotated with frame-by-frame bounding boxes. Each instance undergoes a complete process of being occluded and subsequently reappearing. On this dataset, SKF-Tracker achieved tracking success rates of 89.37% for visible-light videos and 91.93% for infrared videos, representing improvements of 14% and 11.73% over baseline, respectively. In terms of real-time performance, SKF-Tracker maintains high accuracy while achieving 31.25 FPS, significantly faster than deep neural network-based trackers, and does not consume NPU resources on the edge side. This provides a practical solution for stable long-term surveillance missions by drones in complex environments.

4. This dissertation designs and implements the image processing subsystem of an airborne intelligent electro-optical system using multiple edge computing modules. It integrates multi-spectral data acquisition, embedded intelligent processing, and real-time communication. The system uses a highly integrated embedded edge computing module as its core processor, supporting the acquisition and processing of multiple visible-light and infrared video streams. In terms of software, a highly modular and extensible embedded software framework was designed. This framework employs various design patterns to achieve efficient collaboration among data acquisition, processing, and communication services. The integration of lock-free ring buffers, memory pools, and condition variable enhances the system's real-time performance. The system establishes bidirectional data channels with a ground control station via multiple communication protocols (TCP/IP, UDP, HTTP, serial port), establishing a "human-in-the-loop" hybrid collaborative architecture. This solution was integrated and validated through field flight tests on a DJI M350 RTK drone, proving its end-to-end operational capability from data acquisition and real-time processing to information interaction. This system not only provides a reliable engineering validation platform for the algorithmic development but also offers concrete technical references for subsequent algorithm deployment and system integration.

KEY WORDS: Unmanned Aerial Vehicles; Electro-Optical System; Edge Computing; Object Detection and Tracking

TYPE OF DISSERTATION: Application Research

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攻读学位期间取得的研究成果

I. 学术论文

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II. 专利

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